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Cross-certification for secure binding of cryptographic systems

Abstract

Methods and systems for cross-certification to bind together two cryptographic systems are disclosed. One method includes receiving, from a first certificate authority at a second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the first certificate authority. The method includes generating a cross certificate and embedding the hash value as a certificate extension within the cross certificate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority as a continuation-in part application to U.S. patent application Ser. No. 17/344,463, filed on Jun. 10, 2021, which claims priority to U.S. Provisional Patent Application No. 63/037,858, filed on Jun. 11, 2020, the disclosures of each of which are hereby incorporated by reference in their entireties.

BACKGROUND

- (1) In cryptography, a certificate authority or certification authority (CA) is an entity that issues digital certificates. A digital certificate certifies the ownership of a public key by the named subject of the certificate. This allows others (relying parties) to rely upon signatures or on assertions made about the private key that corresponds to the certified public key. A CA acts as a trusted third party—trusted both by the subject (owner) of the certificate and by the party relying upon the certificate. The format of these certificates is specified by the X.509 standard.
- (2) In this context, a root certificate is a public key certificate that identifies a root certificate authority (CA). Root certificates are self-signed and form the basis of an X.509-based public key infrastructure (PKI). Additionally, a certificate authority can issue multiple certificates in the form of a tree structure. A root certificate is the top-most certificate of the tree, the private key of which is used to “sign” other certificates. All certificates signed by the root certificate, with the “CA” field set to true, inherit the trustworthiness of the root certificate. Such a certificate is called an intermediate certificate or subordinate CA certificate. Certificates further down the tree also depend on the trustworthiness of the intermediates.
- (3) Cross certificates generally correspond to a certification of a root key of a Certificate Authority (CA) that is signed by another Certificate Authority (CA). This provides trust between the two certificate authorities in the direction of the certification. For example, a first CA sends its public key to a second CA, and the second CA generates a cross certificate by signing that public key with its own private key. Thereon, any certificates issued by the first CA will now be trusted by the second CA, or entities that trust that second CA. This may not only occur with respect to root certificates, but intermediate certificates as well.
- (4) For trust to be bidirectional, a cross certificate pair is issued, where the second CA also sends its public key to the first CA and the first CA generates a cross certificate by signing the second CA's public key with its own private key. Once this two-way key exchange and signing process is complete, the two CAs can trust each other.
- (5) However, if one of the two root keys is compromised, there is no convenient way for the other

root to detect tampering. In this instance, the second (non-compromised) root would trust any forged certificates from the first root.

(6) Traditionally, compromise of root certificates is of somewhat low risk due to the complex computational demands required to compromise a certificate constructed using RSA or elliptic curve key generation techniques. However, with improvements in computing technology, including the advent of quantum computing, the time required to compromise a certificate is expected to drastically decrease. Accordingly, alternative keying structures are being investigated. However, those alternative keying structures may need to be cross-compatible with existing X.509 standard systems. Accordingly, improvements in the way in which CAs may trust each other, and detect tampering, are desired.

SUMMARY

(7) In general, the present disclosure relates to cross-certification between two different cryptographic systems for purposes of securely binding the systems together. In some aspects, cross certificates may be generated that include attributes of the respective cryptographic systems and are securely signed across those attributes by the respective certificate authorities issuing the cross certificates, to provide a trusted binding between the two cryptographic systems.

(8) In a first aspect, a computer-implemented method includes receiving, from a first certificate authority at a second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the first certificate authority. The method further includes, in response to receipt of the cross certificate request, calculating a signature at the second certificate authority based at least in part on the attribute. The method also includes calculating a hash value based on a combination of the signature calculated at the second certificate authority and a key associated with the second certificate authority; and generating a cross certificate and embedding the hash value as a certificate extension within the cross certificate.

(9) In a second aspect, a system includes a first root certificate authority implementing a first digital signature scheme, the first root certificate authority being communicatively connected to a second root certificate authority implementing a second digital signature scheme different from the first digital signature scheme. The first root certificate authority executes instructions to: receive, from the second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the second certificate authority; in response to receipt of the cross certificate request, calculate a signature based at least in part on the attribute; calculate a hash value based on a combination of the signature and a key associated with the first certificate authority; and generate a cross certificate and embedding the hash value as a certificate extension within the cross certificate.

(10) In a third aspect, a computer-implemented method includes receiving, from a first certificate authority at a second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the first certificate authority. The method further includes, in response to receipt of the cross certificate request, calculating a signature at the second certificate authority based at least in part on the attribute. The method also includes calculating a hash value based on a combination of the signature calculated at the second certificate authority and a key associated with the second certificate authority; and generating a cross certificate and embedding the hash value as a certificate extension within the cross certificate. The method further includes receiving, from the second certificate authority at the first certificate authority, a second cross certificate request, the second cross certificate request including a second attribute unique to the second certificate authority. The method also includes, in response to receipt of the second cross certificate request, calculating a second signature at the first certificate authority based at least in part on the second attribute. The method includes calculating a second hash value based on a combination of the second signature calculated at the first certificate authority and a key associated with the first certificate authority, and generating a second cross certificate and embedding the second hash value as a certificate extension within the second cross certificate.

- (11) In a further aspect, a computer-implemented method includes receiving, from a first certificate authority at a second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the first certificate authority. The method also includes generating, at the second certificate authority, a cross certificate and embedding the attribute as a certificate extension within the cross certificate, and signing the cross certificate with a key associated with the second certificate authority.
- (12) In a still further aspect, a system includes a first root certificate authority implementing a first digital signature scheme, the first root certificate authority being communicatively connected to a second root certificate authority implementing a second digital signature scheme different from the first digital signature scheme. The first root certificate authority executes instructions to: calculate a hash of a signature of a certificate associated with the first certificate authority; generate, at the first root certificate authority, a cross certificate request including an attribute unique to the first certificate authority, the attribute including the hash; and receive a cross certificate including the hash value as a certificate extension within the cross certificate.
- (13) In yet another aspect, a computer-implemented method includes, at a first certificate authority: receiving, from a second certificate authority at the first certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the second certificate authority; generating, at the first certificate authority, a first cross certificate and embedding the attribute as a certificate extension within the first cross certificate; and signing the first cross certificate with a key associated with the first certificate authority. The method further includes, at the second certificate authority: receiving, from the first certificate authority at a second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the first certificate authority; generating, at the second certificate authority, a cross certificate and embedding the attribute as a certificate extension within the cross certificate; and signing the cross certificate with a key associated with the second certificate authority.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example environment including a plurality of digital signature systems in which aspects of the present disclosure may be implemented.
- (2) FIG. 2 illustrates an example computing device with which aspects of the present disclosure may be implemented.
- (3) FIG. 3 illustrates an example parallel cross certification arrangement implemented according to an example embodiment of the present disclosure.
- (4) FIG. 4 is a flowchart of a method of binding digital signature systems using cross certificates, according to an example implementation.
- (5) FIG. 5 is a flowchart of a method of binding two PKI systems at root certificates using cross certificates, according to an example embodiment.
- (6) FIG. 6 is a flowchart of a method of binding two PKI systems at intermediate certificates using cross certificates, according to an example embodiment.
- (7) FIG. 7 is a flowchart of a method of establishing trust at end entities, according to an example embodiment.
- (8) FIG. 8 is a flowchart of a method of validating a binding between two PKI systems at root certificates, according to an example embodiment.
- (9) FIG. 9 is a flowchart of a method of validating a binding between two PKI systems at intermediate certificates, according to an example embodiment.
- (10) FIG. 10 is a flowchart of a method of performing end entity path validation across two PKI systems, according to an example embodiment.

- (11) FIG. 11 illustrates an example enhanced verification process useable in the context of the end entity path validation of FIG. 10.
- (12) FIG. 12 is a flowchart of a method of binding digital signature systems using cross certificates, according to a further example implementation.
- (13) FIG. 13 is a flowchart of a method of binding two PKI systems at root certificates using cross certificates, according to a further example embodiment.
- (14) FIG. 14 is a flowchart of a method of binding two PKI systems at intermediate certificates using cross certificates, according to a further example embodiment.

DETAILED DESCRIPTION

- (15) As briefly described above, embodiments of the present invention are directed to methods and systems for cross certification of parallel PKI certificates. In example embodiments, at the time of cross certification, each root certificate authority will submit a piece of information to be signed using a root key of the other root certificate authority. That signed piece of information can then be embedded to the cross certificates. This can be, for example, submitted as an attribute in a Certificate Request. In some example embodiments, a hash of the certificate of the root may be submitted.
- (16) As noted above, the methods and systems described herein are particularly useful in managing secure binding of two different cryptographic systems, such as an existing system using RSA or elliptic curve key generation, and a post-quantum key generation technique. The methods and systems described herein have further advantages in that a given entity need not maintain and/or carry certificates from both coexisting cryptographic systems, thereby reducing the amount of overhead in terms of bandwidth and/or processing required to select the appropriate key for validation purposes.
- (17) I. System and Environment for Cross-Certification of Cryptographic Systems
- (18) Referring first to FIG. 1, an example environment 10 in which aspects of the present disclosure may be implemented is shown. In the example environment 10, of a plurality of root nodes 100, 102 may be utilized. Each root node may implement a root certificate authority (CA). In other words, each root certificate authority may issue a root certificate useable within a different public key infrastructure (PKI). However, each root certificate authority may be implemented to use a different digital signature scheme. For example, a first root certificate authority at root node 100 may utilize a RSA scheme, while a second root certificate authority at retail node 102 may utilize a different digital signature scheme, such as a SPHINCS+ or Dilithium (next generation) scheme.
- (19) In this example, each root node 100, 102 may be associated with one or more intermediate nodes. As shown, root node 100 is associated with one or more intermediate nodes 110, and root node 102 is associated with one or more intermediate nodes 112. Intermediate nodes 110, 112 may act as intermediate certificate authorities, and may receive intermediate certificates issued by (and signed by) the root certificate authorities. Additionally, intermediate nodes 110, 112 may, if acting as intermediate certificate authorities, issue certificates to one or more end entities 114a-n, 116a-n.
- (20) In this context, it is noted that a certificate generated by an intermediate node will be signed by a root node, indicating validity of the intermediate node certificate. Similarly, end entity certificates may be signed by the intermediate node, acting as intermediate certificate authority and using an intermediate certificate. Accordingly, starting at an end entity, trust chains may form back to the root node, via the end entity certificate, intermediate certificate, and root certificate. In this arrangement, the intermediate nodes and end entity nodes may elect to use a same digital signature scheme as a root node. In other words, in the example shown the intermediate node 110 and end entities 114a-n will use a common digital signature scheme with root node 100, while intermediate node 112 and end entities 116a-n are able to use a common digital signature scheme with root node 102.
- (21) As noted above, it may be that a particular enterprise or group of entities wishes to migrate

from a first digital signature scheme (e.g., a first PKI system) to a second digital signature scheme (e.g., a second PKI system), while remaining within a backwards-compatible, general public key infrastructure. However this may be done gradually, such that certain nodes use an older digital signature scheme while other nodes using a newer digital signature scheme. Or, there may be other reasons to maintain two different digital signature schemes in parallel. In the example network **10** illustrated, validation of one end entity certificate or intermediate certificate with its root would not necessarily mean that another certificate is valid in the other digital signature scheme. Accordingly, methods and systems for binding together to digital signature schemes using cross certificates are described herein.

(22) FIG. 2 illustrates an example computing device **200** with which aspects of the present disclosure may be implemented. The example computing device **200** may be used to implement any individual entity or group of entities, such as the nodes **100**, **102**, **110**, **112**, **114a-n**, and **116a-n** described above in conjunction with FIG. 1.

(23) In the example of FIG. 2, the computing device **200** includes a memory **202**, a processing system **204**, a secondary storage device **206**, a network interface card **208**, a video interface **210**, a display unit **212**, an external component interface **214**, and a communication medium **216**. The memory **202** includes one or more computer storage media capable of storing data and/or instructions. In different embodiments, the memory **202** is implemented in different ways. For example, the memory **202** can be implemented using various types of computer storage media, and generally includes at least some tangible media. In some embodiments, the memory **202** is implemented using entirely non-transitory media.

(24) The processing system **204** includes one or more processing units, or programmable circuits. A processing unit is a physical device or article of manufacture comprising one or more integrated circuits that selectively execute software instructions. In various embodiments, the processing system **204** is implemented in various ways. For example, the processing system **204** can be implemented as one or more physical or logical processing cores. In another example, the processing system **204** can include one or more separate microprocessors. In yet another example embodiment, the processing system **204** can include an application-specific integrated circuit (ASIC) that provides specific functionality. In yet another example, the processing system **204** provides specific functionality by using an ASIC and by executing computer-executable instructions.

(25) The secondary storage device **206** includes one or more computer storage media. The secondary storage device **206** stores data and software instructions not directly accessible by the processing system **204**. In other words, the processing system **204** performs an I/O operation to retrieve data and/or software instructions from the secondary storage device **206**. In various embodiments, the secondary storage device **206** includes various types of computer storage media. For example, the secondary storage device **206** can include one or more magnetic disks, magnetic tape drives, optical discs, solid-state memory devices, and/or other types of tangible computer storage media.

(26) The network interface card **208** enables the computing device **200** to send data to and receive data from a communication network. In different embodiments, the network interface card **208** is implemented in different ways. For example, the network interface card **208** can be implemented as an Ethernet interface, a token-ring network interface, a fiber optic network interface, a wireless network interface (e.g., WiFi, WiMax, etc.), or another type of network interface.

(27) In optional embodiments where included in the computing device **200**, the video interface **210** enables the computing device **200** to output video information to the display unit **212**. The display unit **212** can be various types of devices for displaying video information, such as an LCD display panel, a plasma screen display panel, a touch-sensitive display panel, an LED screen, a cathode-ray tube display, or a projector. The video interface **210** can communicate with the display unit **212** in various ways, such as via a Universal Serial Bus (USB) connector, a VGA connector, a digital

visual interface (DVI) connector, an S-Video connector, a High-Definition Multimedia Interface (HDMI) interface, or a DisplayPort connector.

(28) The external component interface **214** enables the computing device **200** to communicate with external devices. For example, the external component interface **214** can be a USB interface, a FireWire interface, a serial port interface, a parallel port interface, a PS/2 interface, and/or another type of interface that enables the computing device **200** to communicate with external devices. In various embodiments, the external component interface **214** enables the computing device **200** to communicate with various external components, such as external storage devices, input devices, speakers, modems, media player docks, other computing devices, scanners, digital cameras, and fingerprint readers.

(29) The communication medium **216** facilitates communication among the hardware components of the computing device **200**. The communications medium **216** facilitates communication among the memory **202**, the processing system **204**, the secondary storage device **206**, the network interface card **208**, the video interface **210**, and the external component interface **214**. The communications medium **216** can be implemented in various ways. For example, the communications medium **216** can include a PCI bus, a PCI Express bus, an accelerated graphics port (AGP) bus, a serial Advanced Technology Attachment (ATA) interconnect, a parallel ATA interconnect, a Fiber Channel interconnect, a USB bus, a Small Computing system Interface (SCSI) interface, or another type of communications medium.

(30) The memory **202** stores various types of data and/or software instructions. The memory **202** stores a Basic Input/Output System (BIOS) **218** and an operating system **220**. The BIOS **218** includes a set of computer-executable instructions that, when executed by the processing system **204**, cause the computing device **200** to boot up. The operating system **220** includes a set of computer-executable instructions that, when executed by the processing system **204**, cause the computing device **200** to provide an operating system that coordinates the activities and sharing of resources of the computing device **200**. Furthermore, the memory **202** stores application software **222**. The application software **222** includes computer-executable instructions, that when executed by the processing system **204**, cause the computing device **200** to provide one or more applications. The memory **202** also stores program data **224**. The program data **224** is data used by programs that execute on the computing device **200**.

(31) Although particular features are discussed herein as included within an electronic computing device **200**, it is recognized that in certain embodiments not all such components or features may be included within a computing device executing according to the methods and systems of the present disclosure. Furthermore, different types of hardware and/or software systems could be incorporated into such an electronic computing device.

(32) In accordance with the present disclosure, the term computer readable media as used herein may include computer storage media and communication media. As used in this document, a computer storage medium is a device or article of manufacture that stores data and/or computer-executable instructions. Computer storage media may include volatile and nonvolatile, removable and non-removable devices or articles of manufacture implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. By way of example, and not limitation, computer storage media may include dynamic random access memory (DRAM), double data rate synchronous dynamic random access memory (DDR SDRAM), reduced latency DRAM, DDR2 SDRAM, DDR3 SDRAM, solid state memory, read-only memory (ROM), electrically-erasable programmable ROM, optical discs (e.g., CD-ROMs, DVDs, etc.), magnetic disks (e.g., hard disks, floppy disks, etc.), magnetic tapes, and other types of devices and/or articles of manufacture that store data. Communication media may be embodied by computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” may describe a signal that has

one or more characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, radio frequency (RF), infrared, and other wireless media.

(33) It is noted that, in some embodiments of the computing device **200** of FIG. 2, the computer-readable instructions are stored on devices that include non-transitory media. In particular embodiments, the computer-readable instructions are stored on entirely non-transitory media.

(34) FIG. 3 illustrates an example parallel cross certification arrangement **300** implemented according to an example embodiment of the present disclosure. The example parallel cross certification arrangement **300** represents one possible manner in which certificates may be certified across two different PKI systems.

(35) In the example shown, a first root certificate **302** may be bound to a second root certificate **304** by one or more cross certificates **312**, **314**. In the example shown, the first root certificate **302** contains a key (K.sub.1) **306** generated by the first root certificate authority (e.g., root node **100** of FIG. 1), and the second root certificate **304** contains a key (K.sub.2) **308** generated by the second root certificate authority (e.g., root node **102** of FIG. 1).

(36) In the example shown, the first cross certificate **312** may be generated at a second certificate authority. The first cross certificate **312** is signed by the second root certificate **304**, and contains an attribute of the first root certificate **302**. For example, the attribute may be an attribute unique to the first root certificate **302**, such as a hash of a TBSCertificate component associated with the first root, e.g., including key (K.sub.1) **306** generated by the first root certificate authority as well as other attributes embedded therein. The signature by the second root certificate **304** may be applied across the received attribute. In some cases, the first cross-certificate **312** can store a hash value that is based on a combination of the signature calculated at the second certificate authority and the key (K.sub.2) **308** associated with the second certificate authority. This hash value may be stored in a certificate extension within the cross certificate **312**.

(37) Accordingly, the first cross certificate **312** will include information that is unique to the first root certificate **302** and the second root certificate **304** embedded within the hash value. Therefore, validating the cross-certificate, including validation of the certificate extension in accordance with the below, would also validate the binding between the root certificates **302**, **304**.

(38) Similarly, as shown, a second cross certificate **314** may be generated at a first root certificate authority (e.g., root node **100** of FIG. 1), in response to a cross certificate request from a second root certificate authority (e.g., root node **102** of FIG. 1). In that instance, the request from the second root certificate authority can include an attribute unique to the second root certificate authority, for example a hash of the TBSCertificate component at the second root, e.g., including key (K.sub.2) **308** generated by the second root certificate authority and other attributes associated therewith. In formulating the cross certificate **314**, the first root certificate authority can generate a signature over a set of data including its own attributes and the received unique attribute of the first root certificate authority, and can generate a hash of that signature for inclusion in the cross certificate **314** (e.g., within a certificate extension). In this way, unique information from both root certificate authorities is included in the cross certificate **314**, while reducing the overall information that is included by hashing the signature.

(39) Still referring to FIG. 3, a similar cross certificates scheme may be applied between intermediate certificates **322**, **324**. In the example shown, the first intermediate certificate **322**, referred to as “Sub 1” is signed by the first root certificate **302**, and includes a key (IntK.sub.1) **326**, while the second intermediate certificate **324**, referred to as “Sub 2”, is signed by the second root certificate **304** and includes a key (IntK.sub.2) **328**.

(40) In this example, a first intermediate certificate authority that is associated with first intermediate certificate **322** may issue a cross certificate request to a second intermediate certificate authority associated with the second intermediate certificate **324** (e.g., intermediate nodes, as seen

in FIG. 1). As with the root certificates, the first intermediate certificate authority may include in the cross certificate request information unique to the first intermediate certificate **322**, such as a hash of a TBSCertificate component associated with the first root, e.g., including key (IntK.sub.1) **326**. The second intermediate certificate authority may then generate a cross certificate **332** which includes the unique attribute (e.g., the hash of the TBSCertificate of the first intermediate certificate authority) as well as a signature using the second root certificate **304** across information including that attribute. As above, in some examples the attribute information included in the cross-certificate **332** may be included in a hash of the signature, and may be stored in a certificate extension within the cross certificate **332**. This is allowable because V3 extensions are allowed to extend behavior of X.509 compliant certificates.

(41) Similarly, the second intermediate certificate authority that is associated with second intermediate certificate **324** may issue a cross certificate request to the first intermediate certificate authority. In that case, the second intermediate certificate authority may include in that request information unique to the second intermediate certificate **324**, such as a hash of a TBSCertificate of the second intermediate certificate authority. The first intermediate certificate authority may then issue a cross certificate **334** which includes, in a certificate extension, information representing a signature by the intermediate certificate **322** across TBSCertificate of the first intermediate certificate authority.

(42) In the example shown, two end entity certificates **342**, **344** are shown. Generally, end entities will not be configured to sign other certificates. Accordingly, end entities are not shown as performing cross certification. Rather, in typical cases, end entities may request certification from the alternative PKI system. Because of this, a first end entity may send a certification request to obtain end entity certificate **342**; the first end entity may send a certification request to the second intermediate certificate authority **324**, for certification of its public key. In the certification request to the second intermediate certificate authority **324**, the first end-entity will generate its AlternateHash attribute and add it into the certification request as an attribute. Therefore, when the second intermediate certificate issues **344** it will contain the alternate hash. Accordingly, in the embodiment shown, a single end entity may hold both end entity certificates **342**, **344**; in such a circumstance, the same end entity public key **346** (EEK.sub.1) is used with both end entity certificates. An enhanced verification system would be able to determine, based on a missing or tampered-with attribute that is included in the second end entity certificate **344** (shown as AlternateHash), whether tampering occurred. If the attribute is missing, tampering has occurred. Also, if the end entity key **346** EEK.sub.1 is mismatched between end entity certificates, tampering has also likely occurred. Accordingly, both cryptographic systems must be compromised to compromise the system overall.

(43) II. Methods of Binding Cryptographic Systems Using Cross-Certificates

(44) Referring now to FIGS. **4-7**, methods of binding two digital signature systems are shown using cross-certificates according to various example embodiments. FIG. **4** illustrates a general method, while FIG. **5** illustrates a method performed at root certificate authorities. FIG. **6** illustrates a method that may be performed at intermediate certificate authorities, while FIG. **7** illustrates a method that may be performed at end entity nodes.

(45) FIG. **4** is a flowchart of a method **400** of binding digital signature systems using cross certificates, according to an example implementation. In the example shown, the method **400** includes receiving, from a first certificate authority at a second certificate authority, a cross certificate request (step **402**). In this example, the cross certificate request including an attribute unique to the first certificate authority, such as a hash of a TBSCertificate associated with the first certificate authority. In some instances, the hash may be a predetermined hash method and applied to a TBSCertificate (a “to be signed” certificate) of the first certificate authority, including a key of the first certificate authority.

(46) In response to receipt of the cross certificate request, the method **400** may further include

calculating a signature at the second certificate authority (step **404**). This signature generation is performed over a set of information including the attribute (e.g., the hash of the TBSCertificate). Additionally, the method **400** includes calculating a hash value based on a combination of that signature and a key that is associated with the second certificate authority (step **406**). The second certificate authority can then generate the requested cross certificate, embedding this hash value as a certificate extension into the cross certificate (step **408**). As noted above, the certificate extension may be a V3 certificate extension compatible with X.509 certificate standards.

(47) It is noted that in some example embodiments, cross certificates are desired to be generated by each of the first certificate authority and the second certificate authority. In such cases, steps **402-406** may be performed by both certificate authorities prior to generation of the cross certificates.

(48) Referring now to FIG. 5, a method **500** is shown that implements the general process seen in FIG. 4 in the context of a root certificate authority, such that two different root certificate authorities may cross bind (e.g., root nodes **100, 102** of FIG. 1). Typically, the two root certificate authorities may be root authorities associated with two different PKI systems.

(49) In the example shown at a first root certificate authority will generate a hash of the signature portion of its TBSCertificate (step **502**). This has the effect of reducing the size of the signature, while maintaining a unique identifier of that signature (and therefore of the first root certificate authority). The first root certificate authority will then add this hash value as a new attribute type into a cross-certification request (step **504**). The cross certification request may be for example compliant with the PKCS10 standard for submission of a certification request, as defined in RFC 2986. For example, in some instances, a certificate request could include an attribute having the following construct:

(50) TABLE-US-00001 Attribute ::= SEQUENCE { type AttributeType, values SET OF AttributeValue -- at least one value is required -- } AttributeType ::= OBJECT IDENTIFIER
AttributeValue ::= ANY DEFINED BY type

(51) The method **500** further includes submission of the cross certification request to a second root certificate authority (step **506**).

(52) At the second root certificate authority, the cross certificate will be generated (step **508**). This includes calculating a signature over the received attribute (e.g., hash value), and hashing the result of the signature in combination with a key of the second root certificate authority. The second root certificate authority will embed this hash value as a new certificate extension in the generated a cross certificate (step **510**). An example structure for such a certificate extension including the hash attribute value may be as follows:

(53) TABLE-US-00002 Extension ::= SEQUENCE { extnID OBJECT IDENTIFIER,
critical BOOLEAN DEFAULT FALSE, extnValue OCTET STRING } alternateHash
::= SEQUENCE { extnID <TBD>, critical BOOLEAN DEFAULT
FALSE, hashAttributeValue HashAttributeValue }

(54) Referring to method **500** generally, it is recognized that the above described method results in generation of a single cross certificate by the second root certificate authority. In some examples, the method **500** may be repeated in an opposite direction, with the second root certificate authority issuing a cross certification request to the first root certificate authority, and the first root certificate authority generating a cross certificate using analogous operations.

(55) Illustrating the method **500** in the context of the certificates shown in FIG. 3, root certificate authorities (e.g., root nodes **100, 102** of FIG. 1) may issue root certificates **302, 304**, respectively, and implementing two different digital signature schemes, and accordingly, different PKI systems. In this context, a first root certificate authority (e.g., root node **100**) can issue a cross certificate request to the second root certificate authority (e.g., root node **102**), including an attribute that includes a hash of the signature included in the first root certificate **302** (e.g., a hash of a TBSCertificate including, among other attributes, key (K.sub.1) **306**). The second root certificate authority will then issue the cross certificate **312** including a certificate extension that includes a

hash of a signature using the second root certificate **304** over the received attribute hash. Similarly, the second root certificate authority can issue a cross certificate request to the first root certificate authority including an attribute that includes a hash of the signature included in the second root certificate **304** (e.g., a hash of the TBSCertificate of the second root certificate authority, including, among other elements, key (K.sub.2) **308**). The first root certificate authority will then issue the cross certificate **314** including a certificate extension that includes a hash of a signature using the first root certificate **302** over the received attribute hash.

(56) It is noted that the above cross certification has a number of advantages relative to the existing methods of associating PKI systems. In particular, use of the cross certificates may allow tampering or forgery to be detected when one of the two associated PKI systems has been compromised. Additionally, the inclusion of cross certificates requires only a small enhancement to existing certificate issuance procedures, particularly those in which cross-certificates are already employed. Additionally, as noted below, the added requirements for validation are comparatively minimal. Furthermore, because hash values are used, certificate sizes remain manageable. Still further, while the above method requires enhancement to cross-certificates, existing root certificates remain unmodified.

(57) Referring now to FIG. **6**, a method **600** is shown for cross-certification of intermediate certificates. As illustrated in FIG. **1**, each root certificate authority may issue multiple intermediate certificates to different entities. A root certificate that has been compromised would, if undetected, allow forged intermediate certificates to be created, and cross certified roots and associated entities would not be capable of detecting such tampering. Accordingly, the method **600** provides for cross certification of intermediate certificates, and is premised on cross-certification of root certificates being in place as well. For example, the method **600** may be used in combination with the method **500** described above with respect to FIG. **5**.

(58) In the example shown, the method **600** includes generation of intermediate certificates at root certificate authorities (step **602**). The intermediate certificates may be provided, thereafter, to intermediate certificate authorities.

(59) If binding two PKI systems together at the intermediate certificate level is desired, and if the corresponding root certificates are cross bound as noted above in conjunction with FIG. **5** (as assessed at operation **604**), a certificate request can be generated by one intermediate certificate authority, for transmission to another intermediate certificate authority, specifically requesting a cross certificate (step **606**). As with the root certificate case, the certificate request should include a new attribute of the intermediate certificate to allow that intermediate certificate to be cross certified. In example embodiments, the attribute included can be a hash of a signature component of the TBSCertificate of the requesting entity.

(60) Of course, if the root certificates are not cross certified at operation **604**, the method **600** will fail, denying the ability to cross certify the intermediate certificates.

(61) Once the cross certificate request is submitted to a second intermediate certificate authority, (at step **608**), the second intermediate certificate authority can generate a cross certificate using the attribute (step **610**). In example embodiments, generating the cross certificate includes generating a signature over the attribute, and hashing that signature (e.g., in combination with a key of the second intermediate certificate) to form a unique data attribute. That data attribute can then be embedded as a hash value within an extension of the cross certificate (step **612**), e.g., a V3 extension as noted above.

(62) As with the method **500** of FIG. **5**, although the method **600** depicts a process for creation of one cross certificate, it is recognized that a complementary cross certificate may be created by using the same process, but issuing the certificate request from the other of the intermediate certificate authorities (e.g., from the second intermediate certificate authority to the first intermediate certificate authority).

(63) Illustrating the method **600** in the context of the certificates shown in FIG. **3**, intermediate

certificate authorities (e.g., intermediate nodes **110, 112** of FIG. 1) may receive intermediate certificates. In this context, an intermediate certificate authority (e.g., intermediate node **110**) can issue a cross certificate request to the second intermediate certificate authority (e.g., intermediate node **112**), including an attribute that includes a hash of the signature included in the first intermediate certificate **322** (e.g., a hash of a TBSCertificate associated with the first intermediate certificate authority, including key (IntK.sub.1) **326**). The second intermediate certificate authority can then issue the cross certificate **332** including a certificate extension that includes a hash of a signature using the second intermediate certificate **324** over the received attribute hash. Similarly, the second intermediate certificate authority can issue a cross certificate request to the first intermediate certificate authority including an attribute that includes a hash of the signature included in the second intermediate certificate **324** (e.g., a hash of a TBSCertificate component associated with the second intermediate certificate authority, which can include key (IntK.sub.2) **328**). The first intermediate certificate authority will then issue the cross certificate **334** including a certificate extension that includes a hash of a signature using the first intermediate certificate **322** over the received attribute hash.

(64) It is noted that the above cross certification of intermediate certificates has a number of advantages relative to the existing methods of associating PKI systems, similar to those described above with respect to FIG. 5. In particular, the use of cross certificates allows for detection of tampering or forgery when either the two PKI systems is broken. Furthermore, the additional validation procedures required to assess cross certificates are minimal, and that with prior validation procedures. Still further, certificates remain of similar size due to the use of hash values to exchange unique identifiers between intermediate certificate authorities. Also, intermediate certificates remain unchanged, with only intermediate cross certificates being modified.

(65) In some embodiments, even where cross certificates are used at the root certificate level, use of cross certificates at intermediate certificate authorities may not be used while binding between two PKI systems is maintained. In such cases, the signed TBSCertificate hash could be included within the intermediate certificates themselves. This may be particularly applicable in a system using two certificate authorities which issue certificates to the same intermediate and end entities. However, in this alternative embodiment, enrollment with a certificate authority may become more difficult, since a pre-certificate request would be required.

(66) Referring now to FIG. 7, an example method **700** of establishing trust at end entities is illustrated. In this example, since end entity certificates generally are not permitted to sign other certificates, cross certification of end entity certificates is sub optimal, because it would not be compliant with typical standards for PKI systems. Accordingly, as illustrated, an end entity may become certified by requesting certification of its public key from the certificate authority that issued the certificate to that end entity (step **702**). This may be, for example, an intermediate certificate authority, but could also be a root certificate authority. In that the request for certification, an attribute may be included that appends a hash of the TBSCertificate of the existing end entity, including that end entity key (e.g., a public key, such as EEK.sub.1 **346**).

(67) In example embodiments, a request is submitted to an intermediate certificate authority (step **704**). In this case, an intermediate certificate authority will certify the end entity public key, and include an extension in that certification (step **706**). This can include, for example, including as an extension a hash of the TBSCertificate within the end entity certificate. Such an arrangement can be seen in FIG. 3, within end entity certificate **344**, received from intermediate certificate authority **324**.

(68) Concurrently, the same end entity may receive an end entity certificate **342** from another intermediate certificate authority, e.g., intermediate certificate authority **322**. In other words, end entity certificates **342, 344** may be held by the same end entity, and therefore use the same end entity public key **346** (EEK.sub.1).

(69) In alternative embodiments, the intermediate certificate authority **324** could include an

extension that references a location of the original end entity certificate (e.g., certificate **342**). In such cases, the location of the original end entity certificate would be included in the certificate request issued by the end entity.

(70) Upon determination, at operation **708**, that the end entity is certified, the alternate intermediate certificate authority would return an indication of trust to the end entity (step **710**). However, if the end entity cannot be certified, tampering may have occurred, in which case the method **700** fails and optionally new certificates are issued to one or more of the end entity, the intermediate certificate authority, and/or the root certificate authority.

(71) Referring to method **700** generally, it is noted that in some embodiments, the extension including the hash of the TBS Certificate and/or extension could be included in the original end entity certificate (e.g., within certificate **342**). However, in such cases, the end entity certificate should be generated at the same time as the corresponding (alternate) end entity certificate, and/or would be generated in a manner inconsistent with X.509 standards.

(72) Additionally, in accordance with the method **700**, it is noted that an alternate end entity certificate can be used to detect tampering with the end entity certificate, since that alternate end entity certificate may include the extension referencing the location or having unique information regarding the original end entity certificate. A location of the alternate end entity certificate that includes the relevant extension may be provided using an Authority Information Access (AIA) extension (although other extension types may be used). An example structure useable to store such a hash attribute location is as follows:

(73) TABLE-US-00003 AuthorityInfoAccessSyntax ::= SEQUENCE SIZE (1..MAX) OF
AccessDescription AccessDescription ::= SEQUENCE { accessMethod OBJECT
IDENTIFIER, accessLocation GeneralName } accessLocation Info Access

(74) Additionally, the method **700** requires a single extra certificate generation to add this additional layer of protection. Furthermore, the method **700** is backwards compatible with existing PKI systems, while managing certificate sizes at a minimum added size. Furthermore, the original end entity certificate remains unchanged, with only the alternative end entity certificate (e.g., an added certificate that is used according to a new PKI system (e.g., a post-quantum digital signature scheme, such as a Dilithium) would be modified.

(75) III. Methods of Validating Binding of Cross Certificates

(76) Referring now to FIGS. **8-10**, methods of validating a binding between two different PKI systems are disclosed. FIG. **8** illustrates validation occurring at the root certificate level, while FIG. **9** illustrates validation of a binding via cross certificates at intermediate certificates. FIG. **10** illustrates validation of a binding between two PKI systems at the end entity level.

(77) In general, the methods described herein may be performed using a verifier. The verifier may be, for example, any device or entity useable to validate a certificate chain of that would receive the root certificate for validation. For example, this may occur at the root certificate authority, or may occur at a client device that is requesting verification of the root certificate (e.g., to validate a certificate chain which may require validation of an end entity certificate, intermediate certificate, and root certificate). Alternatively, the verifier can be any verifying entity that can act as a validation authority providing information on behalf of a certificate authority.

(78) As seen in FIG. **8**, a method **800** is shown for validation of a binding between two PKI systems at root certificates. The method **800** may be performed, for example, to ensure that the cross-certification between root certificates described above in conjunction with FIG. **5** remains valid.

(79) In the example shown, the method **800** includes obtaining a root certificate and cross-certificates between the root certificate and the corresponding alternative root certificate at a verifier (step **802**). For example, in the context of FIG. **3**, each of certificates **302**, **304**, **312**, **314** are retrieved. Upon receipt of the root certificate and the cross certificates, the verifier can perform traditional verification of the cross certificates using, for example, validation process is described

in RFC 5280 (step **804**).

(80) In the example shown, the method **800** further includes an enhanced verification process (step **806**). This enhanced verification process will verify the root certificate being verified relative to the cross certificate signed by the alternate root certificate authority, and will also verify the alternate root certificate relative to the cross certificate signed by the original root certificate authority.

(81) Specifically, the enhanced verification process may include computing a hash over the signature value of the certificate being verified (step **808**). When verifying the original root certificate, the hash will be taken over the signature value of that root certificate (which was signed over the TBSCertificate data). The hash algorithm used may be a hash algorithm specified in the extension that is included in the cross certificate signed by the alternate root certificate authority (i.e., the root certificate authority of the alternative PKI system to the root being verified). An example attribute that may be included in the cross certificate for defining a hash algorithm may be structured as follows:

(82) TABLE-US-00004 AlternateHashAttribute :: SEQUENCE { alternateHashAttributeType := <TBD> alternateHashAttributeValue := HashAttributeValue } HashAttributeValue ::= SEQUENCE { hashAlgorithm AlgorithmIdentifier, tbsSignatureHash OCTET STRING -- Hash of TBSCertificate Signature Value }

(83) In the example shown, the computed hash value may then be compared to a hash value included in the cross certificate that was signed by the alternate root certificate authority (step **810**). At operation **812** it is determined whether the hash values are equal. If the hash values are not equal, the enhanced verification will fail. However, if the hash values are equal, at least from the perspective of the root certificate, that certificate is verified.

(84) Notably, the enhanced verification process will also be performed on the alternative root certificate. Specifically, a hash will be taken over the signature value of the alternate root certificate, and compared to a hash value included in the cross certificate that was signed by the original root certificate authority.

(85) Assuming that both sets of hash values match, a successful verification of the root certificate is determined (step **814**).

(86) By way of illustration of the method **800** within the context of the arrangement shown in FIG. 3, in step **804**, cross certificates **312**, **314** would be validated using RFC 5280 processes. Then, during an enhanced verification process, validation of root certificate **302** would involve comparison of hash values between the hash of the signature in root certificate **302** and the alternateHash extension value in the cross-certificate **312**, and validation of root certificate **304** would involve comparison of hash values between the hash of the signature in root certificate **304** and the alternateHash extension value in the cross certificate **314**.

(87) FIG. 9 is a flowchart of a method **900** of validating a binding between two PKI systems at intermediate certificates, according to an example embodiment. Generally, the method **900** is similar to the method **800** of FIG. 8, performed at the intermediate certificate level (assuming that cross-certificates exist between the two intermediate certificates, as shown in FIG. 3).

(88) In the example shown, the method **900** includes a verifier obtaining the root certificates and cross-certificates between two root certificates, as well as intermediate certificates and cross certificates between two PKI systems. For example, in the context of FIG. 3, each of certificates **302**, **304**, **312**, **314**, **322**, **324**, **332**, **334** are retrieved. The verifier may be, for example, any device or entity useable to validate a certificate chain of that would receive the root and intermediate certificates for validation. Upon receipt of the various certificates, the verifier can perform traditional verification of the cross certificates between the two intermediate certificates (e.g., cross certificates **332**, **334**) using, for example, validation process is described in RFC 5280 (step **904**).

(89) Upon successful completion of the validation process of step **904**, an enhanced verification process may be performed (step **906**). The enhanced verification process will check the intermediate certificate that is being verified, and computes a hash value over the signature

included in that intermediate certificate (step **908**). The hash value may be computed using an algorithm specified in the extension that is included in the intermediate cross certificate that was signed by the alternate root certificate authority. In the example shown, the hash value is compared to a hash value in the intermediate cross-certificate AlternateHash extension that was signed by the alternate intermediate certificate authority (step **910**).

(90) At operation **912** it is determined whether the hash values are equal. If the hash values are not equal, the enhanced verification will fail. However, if the hash values are equal, at least from the perspective of the intermediate certificate, that certificate is a verified.

(91) Notably, the enhanced verification process will also be performed on the alternate intermediate certificate. Specifically, a hash will be taken over the signature value of the alternate intermediate certificate, and compared to a hash value included in the cross certificate AlternateHash extension that was signed by the original intermediate certificate authority.

(92) Assuming that both sets of hash values match, a successful verification of the root certificate is determined (step **914**).

(93) By way of illustration of the method **900** within the context of the arrangement shown in FIG. **3**, in step **904**, cross certificates **332**, **334** would be validated using RFC 5280 processes. Then, during an enhanced verification process, validation of intermediate certificate **322** would involve comparison of hash values between the hash of the signature in intermediate certificate **322** and cross certificate **332** AlternateHash extension, and validation of intermediate certificate **324** would involve comparison of hash values between the hash of the signature in intermediate certificate **324** and the cross certificate **334** AlternateHash extension.

(94) Referring now to FIG. **10**, a method **1000** is shown for performing end entity path validation across two PKI systems, according to an example embodiment. The method **1000** may be performed, for example, by a verifier seeking verification of end entity certificates held by a single end entity, e.g., an end entity receiving certificates **342**, **344** based on operation of an end entity certificate issuance process, such as the one described above in connection with FIG. **7**.

(95) In the example shown, the method **1000** includes obtaining root certificates and related cross certificates of the bound PKI systems, as well as intermediate certificates and intermediate cross certificates of those PKI systems, alongside end entity certificates (step **1002**). This may be performed using standard protocols defined in RFC 5280. Within the context of the system **300** of FIG. **3**, this step includes obtaining, at the verifier, each of certificates **302**, **304**, **312**, **314**, **322**, **324**, **332**, **334**, **342**, **344**.

(96) Upon obtaining each of the certificates, the verifier may perform verification processes on the end entity certificates using existing validation routines, such as those defined in RFC 5280 (step **1004**). This involves, for example, verifying the chain of certificates from the end entity certificate to the root certificate within the same PKI system. (e.g., validating the certificate chain **342**, **322**, **302**, or the certificate chain **344**, **324**, **304**).

(97) Upon completion of the validation in step **1004**, an enhanced verification process is performed (step **1006**). Generally, the enhanced verification process verifies the validity of the end-entity certificates **342**, **344** and upstream certificates across the PKI systems. A detailed example of the enhanced verification process is described below in connection with FIG. **11**. Generally, the enhanced verification process can lead to, for example, either failure of verification, or successful verification. Upon successful verification, a successful verification may be output (step **1007**).

(98) Referring to FIG. **11**, additional details regarding an example enhanced verification process are provided. The example enhanced verification process **1006** may be performed within the context of verification described above in FIG. **10**, and may be performed by a verifier entity to validate an end entity having end entity certificates associated with two or more certification systems.

(99) In the example shown, the enhanced verification process **1006** can include, for example, assessing an end entity certificate associated with the end-entity, and identifying an extension

containing a hash over the signature value (step **1008**). This can include, for example, obtaining one or both of the end entity certificates that may be present at the end entity. If, at operation **1010**, the extension is absent from the obtained end entity certificate, the verifier will attempt to find the alternate end entity certificate (step **1012**). Additionally, in some cases, even if the extension is present, the alternate end entity certificate can be identified. This can include, for example, using RFC 5280 mechanisms, such as a local directory configuration or repository. Alternatively, the alternate end entity certificate could be included in a P7C package (PKCS#7 certificate list file) in an Authority Information Access (AIA) extension. In other words, in the example of FIG. 3, if the end entity certificate **344** is initially identified, the extension will be detected, and enhanced verification may proceed. If, however, the end entity certificate **342** is initially identified, various mechanisms will be used to identify end entity certificate **344**, which contains the extension.

(100) At operation **1014** if the end entity certificate that includes the extension and the end entity certificate lacking the extension are not both present, entity path validation will fail. However, if at operation **1010**, the extension is present in the originally-accessed end entity certificate, or if at operation **1014** the alternate end entity certificate is found and contains such an extension, the located certificate(s) may be verified, for example using RFC 5280 path validation processes (step **1016**).

(101) In the example shown, the method further includes computing a hash over the original end entity signature value (step **1018**). That is, the hash is computed over the signature value of the end entity certificate that does not include the extension, e.g., using EEK.sub.1 **346** of certificate **342**, as seen in FIG. 3. This can be performed using a hash algorithm specified in the extension located in the alternate end entity certificate **344**. In such an example, the computed hash value may be compared to a hash value in the alternate end entity certificate that was signed by the alternate root certificate (step **1020**). If, at operation **1022**, the hash values match, the enhanced verification process may complete successfully (e.g., leading to a successful verification output at step **1007** of FIG. 10). However, if the hash values do not match, validation of the end entity path will fail.

(102) I. Alternative Methods of Binding Cryptographic Systems Using Cross-Certificates

(103) Referring now to FIGS. **12-14**, alternative methods of binding cryptographic systems are shown, using cross certificates. Generally speaking, the cross certificates described above in conjunction with FIG. 3 may be formed using either the processes of FIGS. 4-7, above, or the processes described herein at FIGS. **12-14**, in combination with the method of FIG. 7 (in the case of establishing trust at end-entities).

(104) FIG. 12 is a flowchart of a method **1200** of binding digital signature systems using cross certificates, according to an example implementation. In the example shown, the method **1200** includes receiving, e.g., from a first certificate authority at a second certificate authority, a cross certificate request (step **1202**). In this example, the cross certificate request including an attribute unique to the first certificate authority, such as a hash of a TBSCertificate associated with the first certificate authority. In some instances, the hash may be a predetermined hash method and applied to a TBSCertificate (a “to be signed” certificate) of the first certificate authority, including a key of the first certificate authority.

(105) The second certificate authority can generate the requested cross certificate, which includes a certificate extension that includes the attribute (step **1204**). As noted above, the certificate extension may be a V3 certificate extension compatible with X.509 certificate standards. In example embodiments, the cross certificate can be signed by the second certificate authority (step **1206**). This signature may be over at least a set of information including the attribute (e.g., the hash of the TBSCertificate). The signed cross certificate may then be returned to the requesting certificate authority (e.g., in this case, the first certificate authority) (step **1208**).

(106) It is noted that in some example embodiments, cross certificates are desired to be generated by each of the first certificate authority and the second certificate authority. In such cases, steps **1202-1208** may be performed by both certificate authorities prior to generation of the cross

certificates.

(107) Referring now to FIG. 13, a method 1300 is shown that implements the general process seen in FIG. 12 in the context of a root certificate authority, such that two different root certificate authorities may cross bind (e.g., root nodes 100, 102 of FIG. 1). Typically, the two root certificate authorities may be root authorities associated with two different PKI systems.

(108) In the example shown at a first root certificate authority will generate a hash of the signature portion of its TBSCertificate (step 1302). This has the effect of reducing the size of the signature, while maintaining a unique identifier of that signature (and therefore of the first root certificate authority). The first root certificate authority will then add this hash value as a new attribute type into a cross-certification request (step 1304). The cross certification request may be for example compliant with the PKCS10 standard for submission of a certification request, as defined in RFC 2986. For example, in some instances, a certificate request could include an attribute having the following construct:

(109) TABLE-US-00005 Attribute ::= SEQUENCE { type AttributeType, values SET OF AttributeValue -- at least one value is required -- } AttributeType ::= OBJECT IDENTIFIER
AttributeValue ::= ANY DEFINED BY type

(110) The method 500 further includes submission of the cross certification request to a second root certificate authority (step 506).

(111) At the second root certificate authority, the cross certificate will be generated (step 1308). As part of this process, the second root certificate authority will embed the attribute as a new certificate extension in the generated a cross certificate. An example structure for such a certificate extension including the hash attribute value may be as follows:

(112) TABLE-US-00006 Extension ::= SEQUENCE { extnID OBJECT IDENTIFIER, critical BOOLEAN DEFAULT FALSE, extnValue OCTET STRING } alternateHash
::= SEQUENCE { extnID <TBD>, critical BOOLEAN DEFAULT FALSE, hashAttributeValue HashAttributeValue }

(113) In the example shown, the cross certificate may also be signed at the second root certificate authority, for example by signing at least a portion of the cross certificate including the received attribute (e.g., hash value), using a key associated with the second root certificate authority.

(114) Referring to method 1300 generally, as with the method 500 of FIG. 5, it is recognized that the above described method results in generation of a single cross certificate by the second root certificate authority. In some examples, the method 1300 may be repeated in an opposite direction, with the second root certificate authority issuing a cross certification request to the first root certificate authority, and the first root certificate authority generating a cross certificate using analogous operations.

(115) Illustrating the method 1300 in the context of the certificates shown in FIG. 3, root certificate authorities (e.g., root nodes 100, 102 of FIG. 1) may issue root certificates 302, 304, respectively, and implementing two different digital signature schemes, and accordingly, different PKI systems. In this context, a first root certificate authority (e.g., root node 100) can issue a cross certificate request to the second root certificate authority (e.g., root node 102), including an attribute that includes a hash of the signature included in the first root certificate 302 (e.g., a hash of a TBSCertificate). The second root certificate authority will then issue the cross certificate 312 including a certificate extension that includes the received attribute, and can sign that certificate using a key K.sub.2 308. Similarly, the second root certificate authority can issue a cross certificate request to the first root certificate authority including an attribute that includes a hash of the signature included in the second root certificate 304 (e.g., a hash of the TBSCertificate of the second root certificate authority). The first root certificate authority will then issue the cross certificate 314 including a certificate extension that includes the attribute, and can sign that cross certificate 314 using a key of the first root certificate authority, K.sub.1 306.

(116) As with the method described above in FIG. 5, this cross certification has a number of

advantages relative to the existing methods of associating PKI systems. In particular, use of the cross certificates may allow tampering or forgery to be detected when one of the two associated PKI systems has been compromised. Additionally, the inclusion of cross certificates requires only a small enhancement to existing certificate issuance procedures, particularly those in which cross-certificates are already employed. Additionally, as noted below, the added requirements for validation are comparatively minimal. Furthermore, because hash values are used, certificate sizes remain manageable. Still further, while the above method requires enhancement to cross-certificates, existing root certificates remain unmodified.

(117) Referring now to FIG. 14, a method **1400** is shown for cross-certification of intermediate certificates. The method **1400**, similar to method **600** of FIG. 6, provides for cross certification of intermediate certificates, and is premised on cross-certification of root certificates being in place as well. For example, the method **1400** may be used in combination with the method **1300** described above with respect to FIG. 13 (much as method **600** of FIG. 6 is useable with the method **500** of FIG. 5).

(118) In the example shown, the method **1400** includes generation of intermediate certificates at root certificate authorities (step **1402**). The intermediate certificates may be provided, thereafter, to intermediate certificate authorities.

(119) If binding two PKI systems together at the intermediate certificate level is desired, and if the corresponding root certificates are cross bound as noted above in conjunction with FIG. 13 (as assessed at operation **1404**), a certificate request can be generated by one intermediate certificate authority (step **1406**), for transmission to another intermediate certificate authority, specifically requesting a cross certificate (step **1408**). As with the root certificate case, the certificate request should include a new attribute of the intermediate certificate to allow that intermediate certificate to be cross certified. In example embodiments, the attribute included can be a hash of a signature component of the TBSCertificate of the requesting entity.

(120) If the root certificates are not cross certified at operation **1404**, the method **1400** will fail, denying the ability to cross certify the intermediate certificates.

(121) Once the cross certificate request is submitted to a second intermediate certificate authority, (at step **1408**), the second intermediate certificate authority can generate a cross certificate using the attribute (step **1410**). In example embodiments, the method **1400** may further include signing the cross certificate (step **1412**), for example signing over at least the attribute, using a key specific to the second intermediate certificate authority.

(122) As with the method **1300** of FIG. 13, although the method **1400** depicts a process for creation of one cross certificate, it is recognized that a complementary cross certificate may be created by using the same process, but issuing the certificate request from the other of the intermediate certificate authorities (e.g., from the second intermediate certificate authority to the first intermediate certificate authority).

(123) Illustrating the method **1400** in the context of the certificates shown in FIG. 3, intermediate certificate authorities (e.g., intermediate nodes **110**, **112** of FIG. 1) may receive intermediate certificates. In this context, an intermediate certificate authority (e.g., intermediate node **110**) can issue a cross certificate request to the second intermediate certificate authority (e.g., intermediate node **112**), including an attribute that includes a hash of the signature included in the first intermediate certificate **322** (e.g., a hash of a TBSCertificate associated with the first intermediate certificate authority, including key (IntK.sub.1) **326**). The second intermediate certificate authority can then issue the cross certificate **332** including a certificate extension that includes the attribute. The second intermediate certificate authority can sign the certificate, for example using a key associated with that certificate authority (e.g., IntK.sub.2 **328**). Similarly, the second intermediate certificate authority can issue a cross certificate request to the first intermediate certificate authority including an attribute that includes a hash of the signature included in the second intermediate certificate **324** (e.g., a hash of a TBSCertificate component associated with the second intermediate

certificate authority). The first intermediate certificate authority will then issue the cross certificate 334 including a certificate extension that includes the attribute, and can sign this cross certificate 334 using a key associated with the first intermediate certificate authority (e.g., IntK.sub.1 326).

(124) It is noted that the above cross certification of intermediate certificates has a number of advantages relative to the existing methods of associating PKI systems, similar to those described above with respect to FIG. 5. In particular, the use of cross certificates allows for detection of tampering or forgery when either the two PKI systems is broken. Furthermore, the additional validation procedures required to assess cross certificates are minimal, and that with prior validation procedures. Still further, certificates remain of similar size due to the use of hash values to exchange unique identifiers between intermediate certificate authorities. Also, intermediate certificates remain unchanged, with only intermediate cross certificates being modified.

(125) In some embodiments, even where cross certificates are used at the root certificate level, use of cross certificates at intermediate certificate authorities may not be used while binding between two PKI systems is maintained. In such cases, the signed TBSCertificate hash could be included within the intermediate certificates themselves. This may be particularly applicable in a system using two certificate authorities which issue certificates to the same intermediate and end entities. However, in this alternative embodiment, enrollment with a certificate authority may become more difficult, since a pre-certificate request would be required.

(126) Referring generally to FIGS. 12-14, the methods of generating cross-certificates corresponds generally to allow for methods of validation of such cross certificates described above in conjunction with FIGS. 8-10. In particular, a holder of a cross certificate can validate that the cross certificate was signed by another entity, and may regenerate its own hash to compare to the attribute included in the cross certificate (e.g., as seen in FIGS. 8-9). In this example, the same entity is generating the hash of a TBSCertificate for comparison to the attribute included in the cross certificate as originally created that attribute (as was included in the original certificate request), and therefore validation is in some ways more straightforward than described above in conjunction with FIGS. 4-6.

(127) Referring to FIGS. 1-14 generally, in accordance with aspects of the present disclosure, it is noted that the present disclosure has a number of advantages over existing systems. For example, the arrangement described herein binds together two separate PKI systems without requiring significant (or in some cases, any) change to structure of certificates, in particular and root certificates and intermediate certificates. Still further, use of hash values of unique attributes of the certificates allows for a compact addition to cross certificates. Furthermore the extension to the overall procedure required for validating trust in certificates adds little computational overhead. Additionally, this binding can be accomplished within the context of existing X.509 standards, ensuring compatibility between formats of certificates of the bound PKI systems. Other advantages are presented by the present disclosure is well as reflected above and in the appended claims.

(128) Although the present disclosure has been described with reference to particular means, materials and embodiments, from the foregoing description, one skilled in the art can easily ascertain the essential characteristics of the present disclosure and various changes and modifications may be made to adapt the various uses and characteristics without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A computer-implemented method comprising: receiving, from a first certificate authority at a second certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the first certificate authority; generating, at the second certificate authority, a cross certificate and embedding the attribute unique to the first certificate authority as a certificate extension within the cross certificate; signing the cross certificate with a key associated with the

second certificate authority; sending, from the second certificate authority to the first certificate authority, a second cross certificate request including an attribute unique to the second certificate authority; and receiving a second cross certificate having embedded therein the attribute unique to the second certificate authority as a certificate extension within the second cross certificate, the second cross certificate being signed with a key associated with the first certificate authority.

2. The computer-implemented method of claim 1, wherein the attribute comprises a hash of a signature of a certificate associated with the first certificate authority.
3. The computer-implemented method of claim 2, wherein signing the cross certificate includes calculating a signature at the second certificate authority using the attribute received as part of the cross certificate request.
4. The computer-implemented method of claim 1, further comprising: at the first certificate authority, validating the cross certificate by: recalculating, after the cross certificate was created, a hash of a signature of the certificate associated with the first certificate authority; and comparing the recalculated hash to the attribute included in the cross certificate.
5. The computer-implemented method of claim 1, wherein the first certificate authority and the second certificate authority comprise root certificate authorities.
6. The computer-implemented method of claim 1, wherein the first certificate authority and the second certificate authority comprise intermediate certificate authorities.
7. The computer-implemented method of claim 1, further comprising: receiving, from the second certificate authority at the first certificate authority, the second cross certificate request; generating, at the first certificate authority, the second cross certificate and embedding the attribute unique to the second certificate authority as the certificate extension within the second cross certificate; and signing the second cross certificate with the key associated with the first certificate authority.
8. The computer-implemented method of claim 1, wherein the first certificate authority uses a first key generation algorithm and the second certificate authority utilizes a second key generation algorithm different from the first key generation algorithm.
9. The computer-implemented method of claim 1, further comprising: requesting certification of a public key of an end entity at the first certificate authority, the first certificate authority being an issuing certificate authority of the public key of the end entity; and issuing a certification request from the first certificate authority to the second certificate authority.
10. The computer-implemented method of claim 9, further comprising: certifying the public key of the end entity at the second certificate authority.
11. The computer-implemented method of claim 1, further comprising: issuing a first end entity certificate to an end entity, the first end entity certificate being associated with the first certificate authority; issuing a second end entity certificate to the end entity, the second end entity certificate being associated with the second certificate authority, the second end entity certificate including an extension that includes a hash of a signature of the second end entity certificate signed by the second certificate authority.
12. The computer-implemented method of claim 11, further comprising: at a verifier, obtaining the first end entity certificate and the second end entity certificate; verifying the first end entity certificate with the first certificate authority; computing a hash of a signature of the first end entity certificate; and comparing the hash of the signature of the first end entity certificate to the hash of a signature of the second end entity certificate to validate the end entity with both the first certificate authority and the second certificate authority.
13. A system comprising: a first root certificate authority implementing a first digital signature scheme; a second root certificate authority implementing a second digital signature scheme different from the first digital signature scheme and communicatively connected to the first root certificate authority; the first root certificate authority executing instructions to: calculate a hash of a signature of a certificate associated with the first root certificate authority; generate, at the first root certificate authority, a cross certificate request including an attribute unique to the first root

certificate authority, the attribute including the hash; receive a cross certificate including the hash as a certificate extension within the cross certificate; and the second root certificate authority executing instructions to: calculate a hash of a signature of a second certificate associated with the second certificate authority; generate, at the second root certificate authority, a cross certificate request including a second attribute unique to the second root certificate authority, the second attribute including the hash of the signature of the second certificate; and receive a second cross certificate including the second hash as a certificate extension within the second cross certificate.

14. The system of claim 13, wherein the first root certificate authority further executes instructions to validate the cross certificate by: recalculating, after the cross certificate was created, a hash of the signature of the certificate associated with the first certificate authority; and comparing the recalculated hash to the attribute included in the cross certificate.

15. The system of claim 14, wherein the first certificate authority is implemented at a first node and the second root certificate authority is implemented at a second node communicatively coupled to the first node.

16. The system of claim 15, further comprising a first intermediate certificate authority and a second intermediate certificate authority, wherein the first intermediate certificate authority is configured to issue a first intermediate certificate signed by the first root certificate authority and the second intermediate certificate authority is configured to issue a second intermediate certificate signed by the second root certificate authority.

17. The system of claim 16, wherein the second intermediate certificate authority is configured to issue a second intermediate cross certificate complementary to the first intermediate cross certificate.

18. A computer-implemented method comprising: at a first certificate authority: receiving, from a second certificate authority at the first certificate authority, a cross certificate request, the cross certificate request including an attribute unique to the second certificate authority; generating, at the first certificate authority, a first cross certificate and embedding the attribute unique to the second certificate authority as a certificate extension within the first cross certificate; and signing the first cross certificate with a key associated with the first certificate authority; and at the second certificate authority: receiving, from the first certificate authority, a second cross certificate request, the second cross certificate request including an attribute unique to the first certificate authority; generating, at the second certificate authority, a second cross certificate and embedding the attribute unique to the first certificate authority as a certificate extension within the second cross certificate; and signing the second cross certificate with a key associated with the second certificate authority.

19. The computer-implemented method of claim 18, further comprising: at the first certificate authority, validating the first cross certificate by: recalculating, after the first cross certificate was created, a hash of the signature of the certificate associated with the first certificate authority; and comparing the recalculated hash to the attribute unique to the second certificate authority included in the first cross certificate; and at the second certificate authority, validating the second cross certificate by: recalculating, after the second cross certificate was created, a second hash of the signature of the certificate associated with the second certificate authority; and comparing the recalculated second hash to the attribute unique to the first certificate authority included in the second cross certificate.
